# Swapnil Meshram

+91 88306 88493 | swapnil.mesh24@gmail.com | LinkedIn | Portfolio

## Education

**Master of Science, Robotics and Autonomous Systems (Electrical Engineering)** **December 2024**
*Arizona State University, Tempe, Arizona* GPA: 3.71/4.00

**Bachelor of Engineering, Electronics Engineering** **May 2020**
*K. K. Wagh Institute of Engineering Education and Research* Nashik, India

## Experience

**Arizona State University: Graduate Research Assistant** **January 2025 - February 2026**
*Miniaturized and Advanced Power Electronics Laboratory* *Tempe, Arizona*

- Engineered semiconductor capacitance extraction tool to reconstruct C-V characteristics from datasheet images and apply the trapezoidal rule to calculate equivalent capacitance values while handling zero-voltage boundary conditions.
- Designed human-in-the-loop computer vision pipeline using binary image thresholding, centroid-based seed refinement and 8-connected Breadth-First Search (BFS) pathfinding to trace complex and highly nonlinear capacitance curves.
- Developed pixel-to-physical coordinate mapping system supporting both linear and logarithmic X/Y axes to convert digitized datasheet coordinates into voltage and capacitance values.

**Arizona State University: Graduate Research Aide** **February 2023 - September 2023**
*Miniaturized and Advanced Power Electronics Laboratory* *Tempe, Arizona*

- Engineered high-frequency planar transformer printed circuit boards (PCBs) that achieved 26 kV and material-enabled 35 kV medium-voltage isolation using 7 kV/mil polyimide for solar thermal applications.
- Co-developed planar transformer PCB designs with Ph.D. researchers for U.S. Department of Energy grant (DE-EE0010239) supporting high-efficiency and low-cost renewable energy integration.
- Optimized complex PCB stack-ups (2-14 layers) to reduce AC copper losses and mitigated fringing electric fields to maintain robust medium-voltage isolation.

**Aerospace Engineers Private Limited: Founding Electrical Engineer** **February 2021 - December 2022**
*Autonomous & Undersea Systems Division* *Tamil Nadu, India*

- Spearheaded R&D for maritime autonomous vehicle electrical systems and achieved 15% cost reduction across key projects including the 300-meter-depth-rated Neerakshi Autonomous Underwater Vehicle (AUV).
- Redesigned the AUV's core power distribution system by eliminating critical endurance constraints and reducing internal wiring complexity by 30% to improve system modularity.
- Engineered high-efficiency propulsion drive through improved thermal management to eliminate active cooling requirements and boost endurance by 5% through optimized Electronic Speed Controller (ESC) tuning.

## Projects

**Deep Learning Approaches to Musical Instrument Sound Classification** **January 2024 - April 2024**
*Skills: Python, PyTorch, Librosa, CNNs, LSTMs, Transformers, Audio Feature Engineering* *Tempe, Arizona*

- Developed end-to-end deep learning pipeline for musical instrument classification to achieve 63.75% accuracy with custom CNN trained on MFCC time-frequency representations.
- Benchmarked LSTM, SVM, Transformer, and ResNet-18 architectures by engineering custom Mel spectrogram and MFCC pipelines to evaluate spatial and temporal feature extraction.

**Comparative Analysis of SLAM Algorithms** **January 2024 - April 2024**
*Skills: ROS, Python, RViz, Cartographer, GMapping, RPLIDAR, Sensor Fusion* *Tempe, Arizona*

- Evaluated Cartographer and GMapping algorithms on ROS-enabled robot equipped with RPLIDAR and measured map precision through MSE and MAE analysis using pixel-to-meter conversion against ground-truth dimensions.
- Identified GMapping's superiority in structured environments and Cartographer's strength in complex layouts and proposed depth-camera sensor fusion to resolve 2D LIDAR blind spots for obstacles under 7 inches.

**EdgeVision: User-Defined Object Counting Using Raspberry Pi** **August 2023 - December 2023**
*Skills: Python, PyTorch, OpenCV, Raspberry Pi, Faster R-CNN, CVAT, Embedded ML* *Tempe, Arizona*

- Engineered real-time object detection pipeline on Raspberry Pi 4 using Faster R-CNN ResNet-50 model trained on custom CVAT-annotated dataset to achieve 95.6% accuracy in bright conditions and 82.5% in low-light environments.
- Developed multi-threaded HTTP video streaming server and custom Centroid Tracker with Euclidean distance matching that handles temporary object occlusions to allow user-defined object counting using interactive line segments.

## Skills

**Languages:** Python, C, Embedded C, MATLAB
**Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, OpenCV, NumPy, Pandas, Matplotlib, Seaborn
**Hardware & Simulation:** Altium Designer, KiCAD, Autodesk Eagle, Simulink, Ansys Maxwell, LTspice
**Tools & Platforms:** CVAT, Linux/Ubuntu, Git/GitHub, VS Code, Jupyter, Google Colab